SPE-226889-MS

Numerical Simulation of Three-Dimensional Fracture Propagation During Supercritical Carbon Dioxide Fracturing

Changxin Yang, China Zhenhua Oil Co., Ltd., Beijing, China / Chengdu Northern Petroleum Exploration and Development Technology Co. Ltd., Chengdu, Sichuan, China; Cheng Ma, China Zhenhua Oil Co., Ltd., Beijing, China / Chengdu Northern Petroleum Exploration and Development Technology Co. Ltd., Chengdu, Sichuan, China; Zhaozhong Yang, Southwest Petroleum University, Chengdu, Sichuan, China; Xin Yang, China Zhenhua Oil Co., Ltd., Beijing, China / Chengdu Northern Petroleum Exploration and Development Technology Co. Ltd., Chengdu, Sichuan, China

 DOI 10.2118/226889-MS

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

By establishing a comprehensive thermal-hydro-mechanical coupled numerical model, this research elucidates the fracture propagation behaviors in unconventional reservoirs subjected to supercritical carbon dioxide (SC-CO_2) fracturing. A comparative analysis is conducted to elucidate the discrepancies between conventional hydraulic fracturing and SC-CO_2 fracturing, and the influence of engineering parameters on the fracture geometries during SC-CO_2 fracturing is revealed, providing useful insights for optimizing SC-CO_2 fracturing strategies and catalyzing the sustainable utilization of carbon dioxide.

An integrated numerical model for three-dimensional (3D) fracture propagation during SC-CO_2 fracturing in unconventional reservoirs is established based on the displacement discontinuity method (DDM) and the finite volume method (FVM). This model comprehensively captures the interactions among rock deformation, fluid flow and crack propagation, and incorporates the variations in rheological parameters of SC-CO_2 and water under different pressure and temperature. A mixed fracture propagation criterion and fracture-matrix exchange model are utilized to describe the fracture trajectories and fluid flow between fractures and reservoir matrix, respectively. The proposed numerical model is solved by Newton-Raphson method and Picard method.

Numerical simulations indicate that leak-off induced stress and thermal stress could change the geometries of hydraulic fracture (HF). Moreover, fracture trajectories and bottomhole pressure variations significantly differ between conventional hydraulic fracturing and SC-CO_2 fracturing. Compared with water, fractures created by SC-CO_2 exhibit smaller fracture width and fracture length under the same injection volume. The heightened diffusivity of SC-CO_2 leads to substantial filtrate losses, triggering notable fluctuations in bottom hole pressure and stress interference that differs from those obtained by conventional hydraulic fracturing treatments. Therefore, the optimized cluster spacing for uniform propagation of multi-fracture differs between conventional hydraulic fracturing and SC-CO_2 fracturing. Sensitive analyses show

that large cluster spacing and high injection rate are favorable for increasing the fracture width and promoting uniform growth of multi-fracture under SC-CO_2 fracturing.

Key words: Supercritical carbon dioxide, Fracture propagation, Three-dimensional model, Multiphysics coupling, Numerical simulation

Introduction

In light of rapid economic development and rising living standards, the demand for fossil energy, particularly oil and natural gas, continues to grow significantly. Consequently, efforts to explore and develop unconventional oil and gas resources are being intensified. The integration of horizontal drilling technology with hydraulic fracturing has enabled the commercial development of unconventional resources (Vengosh et al., 2013), significantly transforming the global energy landscape. Water-based fracturing fluids are predominantly utilized in hydraulic fracturing treatments today due to their cost-effectiveness, ease of viscosity modification, availability, and non-flammable nature. Despite their significant and indispensable role in these treatments, concerns regarding excessive water resource consumption, potential environmental impacts, and suboptimal fracture performance have led to the development and adoption of waterless fracturing technologies (Middleton et al., 2014). Furthermore, the issue of carbon dioxide emissions associated with the development and utilization of fossil fuels has garnered increasing attention from numerous countries. In response to the demand for waterless fracturing technologies and the imperative to reduce carbon emissions (Kim et al., 2021), field engineers proposed the implementation of carbon dioxide fracturing technology to enhance reservoir performance. This method was first executed in field operations in the North America in the 1980s (Gupta et al., 2005).

When pressure and temperature exceed 7.38 MPa and 304.1K, respectively, carbon dioxide transitions into a supercritical state (Span and Wagner, 2005) Supercritical carbon dioxide (SC-CO_2) exhibits unique physical and chemical properties, characterized by a density comparable to that of a liquid, a viscosity similar to that of a gas, extremely low surface tension, and high diffusion rates (Wang et al., 2019). As a specialized waterless fracturing technology, supercritical carbon dioxide fracturing provides several significant advantages, including: (1) reducing the fracture pressure of reservoir rocks to facilitate the propagation of artificial fractures (Shafloot et al., 2021); (2) preventing the swelling of clay minerals, thereby avoiding reservoir damage caused by water-sensitive or water-locking effects associated with conventional water-based fracturing fluids (Bennion et al., 2005; Wang et al., 2020); (3) minimizing the waste of water resources and reducing environmental pollution; and (4) enhancing reservoir performance through the adsorption-replacement interaction between CO_2 and CH_4, while also enabling the geological sequestration of CO_2. Consequently, the advancement of supercritical carbon dioxide fracturing technology is crucial for the efficient development of unconventional resources and for carbon utilization and sequestration (Sampath et al., 2017).

Numerous scholars have undertaken extensive research on SC-CO_2 fracturing, focusing on various aspects, including the mechanisms of CO_2-water-rock interactions, the damage mechanisms of rock fractures under SC-CO_2 conditions, and the law of fracture propagation during the fracturing process (Li et al., 2025) In the realm of laboratory experimental research, the primary focus encompasses monitoring temperature and pressure variations during the SC-CO_2 fracturing process, analyzing the characteristics of artificial fractures, and comparing the differences between SC-CO_2 fracturing and other techniques, including hydraulic fracturing and CO_2 fracturing. Laboratory results have validated the advantages of SC-CO_2 fracturing in creating complex artificial fracture networks and improving reservoir stimulation performance (Zhang et al., 2022). However, limitations related to the performance of experimental apparatus and the size of rock samples mean that qualitative insights derived from laboratory studies cannot be directly applied to field engineering practices. As a result, numerical simulation studies pertaining to supercritical carbon dioxide fracturing at the field scale have increasingly emerged as a focal point of research.

The fracture propagation during SC-CO_2 fracturing presents a complex multiphysics challenge, encompassing thermal transfer, fluid flow, and solid deformation interactions. Li et al. (2018) performed numerical simulations to investigate the effects of fluid flow and heat transfer on fracture propagation during SC-CO_2 fracturing. Zhao et al. (2018) established a numerical model for fracture propagation in heterogeneous shale reservoir during CO_2 fracturing, integrating the maximum tensile stress criterion with the Mohr-Coulomb criterion within a finite element framework. Li et al. (2020) established a numerical simulation model for calculating the distribution of temperature and pressure within the artificial fracture during SC-CO_2 fracturing, and the influence of geo-stress, reservoir temperature and CO_2 temperature were analyzed. They found that higher reservoir temperature is good for SC-CO_2 fracturing. Gong et al. (2019) developed an unsteady-state coupling mathematical model for wellbore pressure drop, analyzing the variations in pressure and temperature within the wellbore during the SC-CO_2 fracturing process, which enables the prediction of CO_2 phase changes throughout the operation. He et al. (2015) integrated laboratory experiments with numerical simulations to investigate the effects of injection parameters on SC- CO_2 jet fracturing, concluding that increasing the pressure differential and fluid temperature can improve fracturing efficiency. Yan et al. (2019) utilized the extended finite element method (XFEM) to simulate various stages of SC-O_2 fracturing, revealing that the length and width of the fracture are directly proportional to the injection rate of SC-CO_2. Likewise, He et al. (2020) developed a coupling model for stress, seepage, and temperature in the context of SC-CO_2 fracturing, analyzing the effects of reservoir and treatment parameters on the propagation of a single fracture. The simulation results indicate that higher reservoir temperatures and increased injection rates result in greater fracture length and width. Song et al. (2021) examined the coupling mechanisms of stress and seepage during the fracturing process through comparative numerical simulations, assessing the effectiveness of conventional hydraulic fracturing versus SC-CO_2 fracturing. The results demonstrate that, under identical injection parameters, SC-CO_2 fracturing generates fractures that are longer and narrower than those produced by hydraulic fracturing. Luo et al. (2021) established a 2D fluid loss model for the SC-CO_2 fracturing process, investigating the fluid loss behavior of fractures under both static and dynamic conditions. Zhang et al. (2021) considered the impact of thermal stress arising from temperature differences between the injected fluid and reservoir rock, exploring the fracture initiation mechanisms of SC-CO_2 fracturing in comparison to hydraulic fracturing in dry hot rocks. The findings indicate that thermal stress induced by lower temperatures can decrease fracture pressure, while low-viscosity fracturing fluids can facilitate the creation of complex fracture networks.

Among these studies, various numerical models were developed and utilized to research critical aspects of SC-CO_2 fracturing, including fracture propagation, fluid loss, temperature and pressure distribution, and the assessment of reservoir enhancement effects. However, the mechanisms governing SC-CO_2 fracturing remain insufficiently understood. This lack of clarity is largely attributed to the sensitivity of its thermophysical parameters during the fracturing process to fluctuations in temperature and pressure, which result in intricate interactions of fluid flow, heat transfer, and fracture propagation. Additionally, the pore-elastic stress resulting from the infiltration of fracturing fluid into the reservoir (Liu et al., 2024), along with the thermal stress induced by temperature variations (Ju et al., 2025) should be appropriately integrated into the numerical simulation.

In the following section, a thermal-hydro-mechanical coupled numerical model for simulating SC-CO_2 fracturing, including rock deformation, fluid flow, fracture propagation and multiphysics coupling effects, will be proposed. By comparing with the published analytical and numerical models, the accuracy and effectiveness of the proposed non-planar 3D fracture propagation model are verified. The influence of pore-elastic stress and thermal stress on fracture propagation is thoroughly analyzed, and the differences between

$SC\text{-}CO_2$ fracturing and hydraulic fracturing are compared. Additionally, the effects of various geological and engineering parameters on $SC\text{-}CO_2$ fracturing are analyzed, and concluding remarks are given.

Mathematical Model

Rock Deformation

The 3D DDM method is used to describe rock deformation during the fracturing process. As illustrated in Fig. 1, the HF within a three-dimensional elastic continuum is discretized into constant displacement discontinuity elements. The stress - displacement expression for each fracture element is formulated as follows (Tang et al., 2022):

$$\begin{cases} \tau_L^i = \sum_{j=1}^{N} A_{LL}^{ij} D_L^j + \sum_{j=1}^{N} A_{LH}^{ij} D_H^j + \sum_{j=1}^{N} A_{LN}^{ij} D_N^j \\ \tau_H^i = \sum_{j=1}^{N} A_{HL}^{ij} D_L^j + \sum_{j=1}^{N} A_{HH}^{ij} D_H^j + \sum_{j=1}^{N} A_{HN}^{ij} D_N^j \\ \sigma_N^i = \sum_{j=1}^{N} A_{NL}^{ij} D_L^j + \sum_{j=1}^{N} A_{NH}^{ij} D_H^j + \sum_{j=1}^{N} A_{NN}^{ij} D_N^j \end{cases} \tag{1}$$

where N is the total element number; subscripts L, H, N refer to the strike, dip, and normal directions, respectively; A_{ij} denotes the influence coefficient, which quantifies the stresses at element i by the displacements of element j and can be calculated analytically; D_L, D_H and D_N are the displacement discontinuities, and fracture width w is equal to the normal displacement discontinuity D_N.

Generally, the deformation of HF is governed by the reservoir in-situ stress and the fluid pressure inside the fracture. By reformulating Eq. (1), the fracture width of element i can be calculated by:

$$C_m w_i = p_{f,i} - \sigma_{m,i} \tag{2}$$

where C_m denotes the comprehensive influence coefficient; w is the fracture width; p_f is the fluid pressure; σ_m is the comprehensive stress, which can be evaluated by in-situ stress.

During hydraulic fracturing, fluid leaks off from the fracture surfaces into the reservoir, causing variations in reservoir pore pressure. Based on the assumption that the pressure diffusion length scale in the reservoir is much smaller than the HF size, the stresses induced by spatial pore pressure variation (Δp_m) during fracturing can be evaluated by Dontsov (2021):

$$\Delta\sigma_{p,ij} = \Delta p_m \frac{\alpha(1-2v)}{1-v}\left(\delta_{ij} + \frac{1}{4\pi} I_{,ij}\right) \tag{3}$$

where α is the Biot coefficient; υ is the Poisson's ratio; δ_ij is Dirac delta function. T.

The expression of the Green's function is given by:

$$I(x, y, z) = \int_{x-a}^{x+a}\int_{y-b}^{y+b}\int_{z-c}^{z+c} \frac{dx'dy'dz'}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}} \tag{4}$$

where (x', y', z') are the coordinates of the source poin; (x, y, z) are the coordinates of the field point; a, b, and c denote the half-lengths of the rectangular cuboid element. The derivatives of $I(x, y, z)$ can be found in Dontsov's work (2021).

Given the temperature differential between fracturing fluid and the surrounding formation rock, the resulting thermal stress acting on the fracture surfaces is expressed as (Zhang et al., 2022):

$$I\Delta\sigma_T = \beta \frac{E\alpha_s}{1-v}\left(T_\mathrm{f} - T_\mathrm{m}\right) \tag{5}$$

where α_s is the linear thermal expansion coefficient of rock; E is the Young's modulus; T_f and T_m are the fracturing fluid temperature and reservoir matrix temperature, respectively; β is a constant related to heat transfer (Collin and Rowcliffe, 2000), which is assigned a value of 0.7 in this study (Lin et al., 2024).

Accordingly, by incorporating the leak-off induced poroelastic stresses and the thermal stresses, the the calculation of fracture width formulation can be reformulated as follows:

$$C_m w_i = p_{f,i} - \sigma_{m,i} - \Delta\sigma_{p,i} - \Delta\sigma_{T,i} \tag{6}$$

Fracturing Fluid Flow

According to Poiseuille's law, fracturing fluid flow within the HF is idealized as laminar flow, with leak-off characterized by the Carter's model (Wu et al., 2015). Accordingly, the continuity equation is given by:

$$\frac{\partial\left(\rho_f w\right)}{\partial t} - \nabla\cdot\left(\frac{\rho_f w^3}{12\mu_f}\nabla p_f\right) + \frac{2\rho_f C_{leak}}{\sqrt{t_c-\tau_0}} = \rho_f q_f \tag{7}$$

where ρ_f and μ_f are the fluid density and viscosity, respectively; C_{leak} is the leak-off coefficient; t_c is the current time; τ_0 is the opening time of the fracture element; q_f is the injection source term of HF.

The reservoir is treated as a saturated poroelastic medium, and the governing equation for fluid flow in the reservoir matrix is formulated as:

$$\rho_f \phi_m C_f \frac{\partial p_m}{\partial t} - \nabla\cdot\left(\frac{\rho_f k_m}{u}\nabla p_m\right) = \rho_f q_m \tag{8}$$

where C_f is the fluid compression coefficient; p_m is the pore pressure of the reservoir matrix; ϕ_m and k_m are the reservoir porosity and permeability, respectively; q_m denotes the injection source term of reservoir matrix.

Given the impracticality of precise leak-off coefficient acquisition, the Embedded Discrete Fracture Model (EDFM) (Yang et al., 2025) is implemented to simulate fluid flux between HF and reservoir, which is governed by:

$$\gamma^{fm} = -\gamma^{mf} = \frac{T^{mf}\left(p_m - p_f\right)}{2\mu_f A_f} \tag{9}$$

where T_{mf} denotes the matrix-fracture transmissibility; A_f is the contact area between fracture and matrix elements.

Consequently, the governing equations for fluid flow in the fracture-matrix system during fracturing are given by:

$$\begin{cases} \dfrac{\partial\left(\rho_f w\right)}{\partial t} - \nabla\cdot\left(\dfrac{\rho_f w^3}{12\mu_f}\nabla p_f\right) = \rho_f q_f + \rho_f \gamma^{fm} \\ \rho_f \phi_m C_f \dfrac{\partial p_m}{\partial t} - \nabla\cdot\left(\dfrac{\rho_f k_m}{\mu_f}\nabla p_m\right) = \rho_f \gamma^{mf} \end{cases} \tag{10}$$

Accounting for the impact of competitive propagation among multiple fractures during fracturing in horizontal well, the injection rates vary among different perforation clusters. According to Kirchhoff's law (Wu et al., 2015), the bottomhole pressure and total injection rate are subject to the following conditions:

$$\begin{cases} p_B = p_{in,i} + p_{p,i} + p_{w,i} (i = 1, 2, \cdots, m) \\ q_{all} = q_{f,1} + q_{f,2} + \cdots + q_{f,m} \end{cases} \tag{11}$$

where m is the number of perforation cluster; p_B is the bottomhole pressure; $p_{in,i}$ represents the inlet pressure of the HF at the i-th cluster; $p_{p,i}$ represents the perforation friction pressure drop of the i-th cluster; $p_{w,i}$ represents the wellbore friction pressure drop from the heel of the horizontal well to the i-th cluster. Using the Newton-Raphson iterative method, the dynamic fracturing fluid injection rates for each perforation cluster in the horizontal well and the corresponding bottomhole fluid pressure can be obtained by solving the Equations (11).

Boundary-Condition and Numerical-Simulation Procedure

Considering temperature variations in both the fracturing fluid and the reservoir during hydraulic fracturing, a thermodynamic model for the fracture-matrix system is developed based on Fourier's law and the local thermal equilibrium assumption.

Under the coupled effects of heat conduction and convection, the energy conservation equation for the reservoir matrix is given by:

$$(pc)_m \frac{\partial T_m}{\partial t} - 3\alpha_m T_m \frac{\partial p_m}{\partial t} - \nabla \cdot (\lambda_m \nabla T_m) + \left(\rho_f c_f \mathbf{v}_m\right) \nabla T_m = \psi^{mf} \tag{12a}$$

and

$$\begin{aligned} (\rho c)_m &= \left(1-\phi_m\right)\rho_s c_s + \phi_m \rho_f c_f \\ \lambda_m &= \left(1-\phi_m\right)\lambda_s + \phi_m \lambda_f \\ \alpha_m &= \left(1-\phi_m\right)\alpha_s + \phi_m \alpha_f \end{aligned} \tag{12b}$$

where ρ_s, c_s, λ_s are the density, specific heat capacity and thermal conductivity of formation rock, respectively; ρ_f, c_f, λ_f are the density, specific heat capacity and thermal conductivity of fracturing fluid, respectively; α_s and α_f are the thermal expansion coefficient of rock and fluid, respectively; $\mathbf{v}_m$ is the fluid velocity in the reservoir matrix.

For the temperature field in the HF, neglecting the impact of variations in fluid pressure, the energy conservation equation is expressed as:

$$\rho_f c_f \frac{\partial T_f}{\partial t} - \nabla \cdot \left(\lambda_f \nabla T_f\right) + \left(\rho_f c_f \mathbf{v}_f\right) \nabla T_f = \psi^{fm} + H_f \tag{13}$$

where $\mathbf{v}_m$ denotes the fluid velocity within the HF.

Similarly, the energy exchange between HF and the reservoir matrix is calculated using the EDFM, which can be calculated by:

$$\psi^{fm} = -\psi^{mf} = H\left(v^{fm}\right) v^{fm} e^f + H\left(-v^{fm}\right) v^{fm} e^m + TI\lambda_e\left(T_m - T_f\right) \tag{14}$$

where H $(\cdot)$ is the Heaviside function; v_{fm} is the fluid exchange velocity; e is the thermal energy carried by fluid; TI is the exchange coefficient; λ_e denotes the equivalent thermal conductivity.

In light of temperature and pressure variations within the HF and surrounding reservoir during fracturing, it is necessary to calculate the thermophysical properties of the fracturing fluid, including density, viscosity, specific heat capacity, and thermal conductivity. To reduce model complexity, this study employs the American NIST database (Odabaee et al., 2016) to source the thermophysical properties of water and CO_2 various temperature and pressure conditions.

Boundary-Condition and Numerical-Simulation Procedure

Non-planar 3D model is utilized to simulate the HF propagation, which enables accounting for fracture deflection under the influence of stress interference among different HFs. Assuming HF propagation follows a quasi-static process, a mixed-mode failure criterion incorporating both tensile failure and shear failure is introduced to determine whether fracture propagation occurs. Based on the linear elastic fracture mechanics (LEFM) theory (Zhang and Robert, 2016) [Ref], fracture extension occurs when the equivalent stress intensity factor (K_e) at the fracture tip equals or exceeds the fracture toughness (K_{IC}),

$$K_e \geq K_{IC} \tag{15}$$

The equivalent stress intensity factor (K_e) can be evaluated by:

$$K_e = \cos\frac{\theta}{2}\left[K_I \cos^2\frac{\theta}{2} - \frac{3}{2}K_{II}\sin\theta\right] \tag{16}$$

where K_I is the tensile-mode stress intensity factor; K_{II} is the shear-mode stress intensity factor. The stress intensity factors can be calculated by:

$$K_I = \frac{0.806E\sqrt{\pi}}{4(1-\nu^2)\sqrt{\Delta l}}D_{N,tip}, \quad K_{II} = \frac{0.806E\sqrt{\pi}}{4(1-\nu^2)\sqrt{\Delta l}}D_{L,tip} \tag{17}$$

where Δl denotes the size of the fracture tip element; $D_{L,tip}$ and $D_{N,tip}$ are the shear and normal displacement discontinuities, respectively.

According to the maximum circumferential stress criterion (Erdogan and Sin, 1963), HFs propagate along the orientation of maximum circumferential tensile stress at the fracture tip. In this scenario, the fracture propagation deflection angle (θ) fulfills the following condition:

$$\frac{1}{\sqrt{2\pi r}}\cos\frac{\theta}{2}\left[\frac{1}{2}K_I\sin\theta + \frac{1}{2}K_{II}(3\cos\theta - 1)\right] = 0 \tag{18}$$

By solving the aforementioned equation, the deflection angle during fracture propagation can be accurately determined. It should be noted that within the non-planar 3D model, the deflection of HFs is considered solely within the plane defined by the maximum and minimum horizontal principal stresses, while no deflection occurs in the vertical direction of the HF. Consequently, when evaluating the potential of HF propagation in the vertical direction, the influence of the shear-mode stress intensity factor is excluded from consideration.

Numerical Solution

Non-planar 3D model is utilized to simulate the HF propagation, which enables accounting for fracture deflection under the influence of stress

To solve the THM coupled numerical model for SC-CO_2 fracturing, a staggered iterative strategy is utilized. The 3D DDM is used to calculate fracture widths during the fracturing process, and the finite volume method (FVM) is applied for the computation of the fluid pressure and temperature field.

The fracture width-fluid pressure matrix expression derived from Eq. (6) is given by:

$$\mathbf{C}_\mathrm{m}\mathbf{w} - \mathbf{p}_\mathrm{f} = \boldsymbol{\sigma}_\mathrm{f} \tag{19}$$

The governing equations of fluid flow for the fracture-matrix system are discretized utilizing the FVM, with the unsteady terms treated using the backward Euler finite difference scheme. It is worth noting that the geometric elements used to solve fluid flow within the fracture are consistent with the displacement discontinuity elements, and the reservoir geometric model is discretized using cubic elements.

According to the theory of FVM and EDFM, the numerical computational scheme of the Eq. (7) can be written as follows:

$$
\begin{aligned}
& w_{i,j}^{t}+\frac{\Delta t}{\left(\rho_f\right)_{i,j}^{t}}\left[\frac{\rho_e\left(w_e^t\right)^3}{12\mu_e\Delta x2}+\frac{\rho_w\left(w_w^t\right)^3}{12\mu_w\Delta x2}+\frac{\rho_n\left(w_n^t\right)^3}{12\mu_n\Delta y2}+\frac{\rho_s\left(w_s^t\right)^3}{12\mu_s\Delta y2}\right]\left(p_f\right)_{i,j}^{t}+\frac{\Delta t T_{i,j}^{mf}}{2\left(\mu_f\right)_{i,j}^{t}\left(A_f\right)_{i,j}}\left(p_f\right)_{i,j}^{t} \\
& -\frac{\Delta t}{\left(\rho_f\right)_{i,j}^{t}}\left[\frac{\rho_e\left(w_e^t\right)^3}{12\mu_e\Delta x2}\left(p_f\right)_{i+1,j}^{t}+\frac{\rho_w\left(w_w^t\right)^3}{12\mu\Delta x2}\left(p_f\right)_{i-1,j}^{t}+\frac{\rho_n\left(w_n^t\right)^3}{12\mu\Delta y2}\left(p_f\right)_{i,j+1}^{t}+\frac{\rho_s\left(w_s^t\right)^3}{12\mu\Delta y2}\left(p_f\right)_{i,j-1}^{t}\right] \\
& -\frac{\Delta t T_{i,j}^{mf}\left(p_m\right)^t}{2\left(\mu_f\right)_{i,j}^{t}\left(A_f\right)_{i,j}}=\Delta t\left(q_f\right)_{i,j}^{t}+w_{i,j}^{t-\Delta t}-w_{i}^{t-\Delta t}\left[1-\frac{\left(\rho_f\right)_{i,j}^{t-\Delta t}}{\left(\rho_f\right)_{i,j}^{t}}\right]
\end{aligned}
\tag{20}
$$

where Δt is the time step; Δx is the fracture element length; Δy is the fracture element height.

The fracture widths at the boundaries of the fracture elements are computed using the arithmetic mean, whereas the viscosity and density at these boundaries are calculated using the harmonic mean. The relevant computational expressions are given by:

$$
\begin{cases}
w_{i+1/2,j}=\frac{w_{i,j}+w_{i+1,j}}{2},\ w_{i-1/2,j}=\frac{w_{i,j}+w_{i-1,j}}{2} \\
w_{i,j+1/2}=\frac{w_{i,j}+w_{i,j+1}}{2},\ w_{i,j-1/2}=\frac{w_{i,j}+w_{i,j-1}}{2}
\end{cases}
\tag{21}
$$

$$
\begin{cases}
\rho_{i+1/2,j}=\frac{2\rho_{i,j}\rho_{i+1,j}}{\rho_{i,j}+\rho_{i+1,j}},\ \rho_{i-1/2,j}=\frac{2\rho_{i,j}\rho_{i-1,j}}{\rho_{i,j}+\rho_{i-1,j}} \\
\rho_{i,j+1/2}=\frac{2\rho_{i,j}\rho_{i,j+1}}{\rho_{i,j}+\rho_{i,j+1}},\ \rho_{i,j-1/2}=\frac{2\rho_{i,j}\rho_{i,j-1}}{\rho_{i,j}+\rho_{i,j-1}}
\end{cases}
\tag{22}
$$

$$
\begin{cases}
\mu_{i+1/2,j}=\frac{2\mu_{i,j}\mu_{i+1,j}}{\mu_{i,j}+\mu_{i+1,j}},\ \mu_{i-1/2,j}=\frac{2\mu_{i,j}\mu_{i-1,j}}{\mu_{i,j}+\mu_{i-1,j}} \\
\mu_{i,j+1/2}=\frac{2\mu_{i,j}\mu_{i,j+1}}{\mu_{i,j}+\mu_{i,j+1}},\ \mu_{i,j-1/2}=\frac{2\mu_{i,j}\mu_{i,j-1}}{\mu_{i,j}+\mu_{i,j-1}}
\end{cases}
\tag{23}
$$

Therefore, the final matrix expression can be formulated as:

$$
\mathbf{w}^{\mathrm{t}}+\left(\mathbf{C}_{\mathrm{f}}+\frac{\Delta t\mathbf{T}^{mf}}{\mu\mathbf{A}_{mf}}\right)\mathbf{p}_{\mathrm{f}}^{\mathrm{t}}-\frac{\Delta t\mathbf{T}^{mf}}{\mu\mathbf{A}_{mf}}\mathbf{p}_{\mathrm{m}}^{\mathrm{t}}=\mathbf{S}_{\mathrm{f}}
\tag{24}
$$

By using the same approach to numerically discretize the governing equation, the final matrix expression for fluid flow in the reservoir matrix can be formulated as follows:

$$
\left(\mathbf{C}_{\mathrm{p}}+\frac{\Delta t\mathbf{T}^{mf}}{\mu\phi C_f\mathbf{A}_{mf}}\right)\mathbf{p}_{\mathrm{m}}^{\mathrm{t}}-\frac{\Delta t\mathbf{T}^{mf}}{\mu\phi C_f\mathbf{A}_{mf}}\mathbf{p}_{\mathrm{f}}^{\mathrm{t}}=\mathbf{0}
\tag{25}
$$

Integrating Eqs. (19), (24) and (25), the hydro-mechanical (HM) coupling governing equations for the rock deformation and fluid flow during fracturing are given by:

$$
\begin{bmatrix}
\mathbf{C}_{\mathrm{m}} & -\mathbf{I} & \mathbf{0} \\
\mathbf{I} & \mathbf{C}_{\mathrm{f}}+\mathbf{C}_{\mathrm{fm}} & -\mathbf{C}_{\mathrm{fm}} \\
 & -\mathbf{C}_{\mathrm{mf}} & \mathbf{C}_{\mathrm{p}}+\mathbf{C}_{\mathrm{mf}}
\end{bmatrix}
\begin{bmatrix}
\mathbf{w} \\ \mathbf{p}_{\mathrm{f}} \\ \mathbf{p}_{\mathrm{m}}
\end{bmatrix}
=
\begin{bmatrix}
\boldsymbol{\sigma}_{\mathrm{f}} \\ \mathbf{S}_{\mathrm{f}} \\ \mathbf{0}
\end{bmatrix}
\tag{26}
$$

where $\mathbf{C}$ is the fluid flow coefficient matrix; $\mathbf{I}$ is the unit matrix; $\mathbf{S}_{\mathrm{f}}$ is the source matrix of the artificial fracture.

The energy conservation equations for the fracture-matrix system are also discretized using the FVM, with the specific operational steps are the same as those applied in the discretization of the fluid flow control equations. Notably, to improve the stability of the numerical solution, the upwind scheme is utilized to discretize the convection terms within the energy conservation equation. For fracture element (i, j), the numerical scheme for convection terms is given by:

$$\begin{aligned}
\int\left(\rho_f c_f \mathbf{v}_f\right)\nabla T_f dA &= \Delta y\left(T_f\right)_{i,j}^{t}\max\left(\rho_f c_f v_{i+1/2,j}, 0\right) - \Delta y\left(T_f\right)_{i+1,j}^{t}\max\left(-\rho_f c_f v_{i+1/2,j}, 0\right)\\
&+\Delta y\left(T_f\right)_{i-1,j}^{t}\max\left(\rho_f c_f v_{i-1/2,j}, 0\right) - \Delta y\left(T_f\right)_{i,j}^{t}\max\left(-\rho_f c_f v_{i-1/2,j}, 0\right)\\
&+\Delta x\left(T_f\right)_{i,j}^{t}\max\left(\rho_f c_f v_{i,j+1/2}, 0\right) - \Delta x\left(T_f\right)_{i,j+1}^{t}\max\left(-\rho_f c_f v_{i,j+1/2}, 0\right)\\
&+\Delta x\left(T_f\right)_{i,j-1}^{t}\max\left(\rho_f c_f v_{i,j-1/2}, 0\right) - \Delta x\left(T_f\right)_{i,j}^{t}\max\left(-\rho_f c_f v_{i,j-1/2}, 0\right)
\end{aligned} \tag{27}$$

For matrix element (i, j, k), the numerical scheme for convection terms is given by:

$$\begin{aligned}
&\int\left(\rho_f c_f \mathbf{v}_m\right)\nabla T_m dV =\\
&\Delta Y\Delta Z\left(T_m\right)_{i,j,k}^{t}\max\left(\rho_f c_f v_{i+1/2,j,k}, 0\right) - \Delta Y\Delta Z\left(T_m\right)_{i+1,j,k}^{t}\max\left(-\rho_f c_f v_{i+1/2,j,k}, 0\right)+\\
&\Delta Y\Delta Z\left(T_m\right)_{i-1,j,k}^{t}\max\left(\rho_f c_f v_{i-1/2,j,k}, 0\right) - \Delta Y\Delta Z\left(T_m\right)_{i,j,k}^{t}\max\left(-\rho_f c_f v_{i-1/2,j,k}, 0\right)+\\
&\Delta X\Delta Z\left(T_m\right)_{i,j,k}^{t}\max\left(\rho_f c_f v_{i,j+1/2,k}, 0\right) - \Delta X\Delta Z\left(T_m\right)_{i,j+1,k}^{t}\max\left(-\rho_f c_f v_{i,j+1/2,k}, 0\right)+\\
&\Delta X\Delta Z\left(T_m\right)_{i,j-1,k}^{t}\max\left(\rho_f c_f v_{i,j-1/2,k}, 0\right) - \Delta X\Delta Z\left(T_m\right)_{i,j,k}^{t}\max\left(-\rho_f c_f v_{i,j-1/2,k}, 0\right)+\\
&\Delta X\Delta Y\left(T_m\right)_{i,j,k}^{t}\max\left(\rho_f c_f v_{i,j,k+1/2}, 0\right) - \Delta X\Delta Y\left(T_m\right)_{i,j,k+1}^{t}\max\left(-\rho_f c_f v_{i,j,k+1/2}, 0\right)+\\
&\Delta X\Delta Y\left(T_m\right)_{i,j,k-1}^{t}\max\left(\rho_f c_f v_{i,j,k-1/2}, 0\right) - \Delta X\Delta Y\left(T_m\right)_{i,j,k}^{t}\max\left(-\rho_f c_f v_{i,j,k-1/2}, 0\right)
\end{aligned} \tag{28}$$

where ΔX, ΔY and ΔZ are the size of matrix elements.

Following numerical discretization, the matrix expression for the numerical calculation of the energy conservation equations in the fracture-matrix system during the fracturing process can be formulated as follows:

$$\begin{bmatrix}\mathbf{U}_{\mathrm{ff}} & \mathbf{U}_{\mathrm{fm}}\\ \mathbf{U}_{\mathrm{mf}} & \mathbf{U}_{\mathrm{mm}}\end{bmatrix}\begin{bmatrix}\mathbf{T}_{\mathrm{f}}\\ \mathbf{T}_{\mathrm{m}}\end{bmatrix}=\begin{bmatrix}\mathbf{Q}_{\mathrm{f}}\\ \mathbf{0}\end{bmatrix} \tag{29}$$

where $\mathbf{U}$ is the heat transfer coefficient matrix; $\mathbf{Q}_{\mathrm{f}}$ is the thermal source matrix of the artificial fracture.

Applying the prescribed parameters along with initial and boundary conditions, the fracture aperture (w), fracture fluid pressure (p_f), matrix pore pressure (p_m), fracture fluid temperature (T_f) and reservoir temperature (T_m) at each time step are calculated by numerically solving the Eqs. (26) and (29) with the Newton-Raphson method. Additionally, the Picard iteration is utilized to improve the stability of the numerical solution. The numerical solution is deemed converged when the following convergence criteria are satisfied.

$$\begin{cases}
\dfrac{1}{N}\sum_{i=1}^{N}\left|\dfrac{w_i^{\mathrm{n}+1}-w_i^{\mathrm{n}}}{w_i^{\mathrm{n}}}\right|<\varepsilon_{tol}, \quad \dfrac{1}{N}\sum_{i=1}^{N}\left|\dfrac{p_{f,i}^{\mathrm{n}+1}-p_{f,i}^{\mathrm{n}}}{p_{f,i}^{\mathrm{n}}}\right|<\varepsilon_{tol}, \quad \dfrac{1}{N}\sum_{i=1}^{N}\left|\dfrac{T_{f,i}^{\mathrm{n}+1}-T_{f,i}^{\mathrm{n}}}{T_{f,i}^{\mathrm{n}}}\right|<\varepsilon_{tol}\\
\dfrac{1}{M}\sum_{i=1}^{N}\left|\dfrac{p_{m,i}^{\mathrm{n}+1}-p_{m,i}^{\mathrm{n}}}{p_{m,i}^{\mathrm{n}}}\right|<\varepsilon_{tol}, \quad \dfrac{1}{M}\sum_{i=1}^{M}\left|\dfrac{T_{m,i}^{\mathrm{n}+1}-T_{m,i}^{\mathrm{n}}}{T_{m,i}^{\mathrm{n}}}\right|<\varepsilon_{tol}
\end{cases} \tag{30}$$

where $\varepsilon_{\mathrm{tol}} = 1.0\times10^{-4}$ defines convergence tolerance; N and M denote the counts of fracture and matrix elements, respectively.

The computational framework of the THM coupled model during SC-CO_2 fracture is illustrated in figure 1.

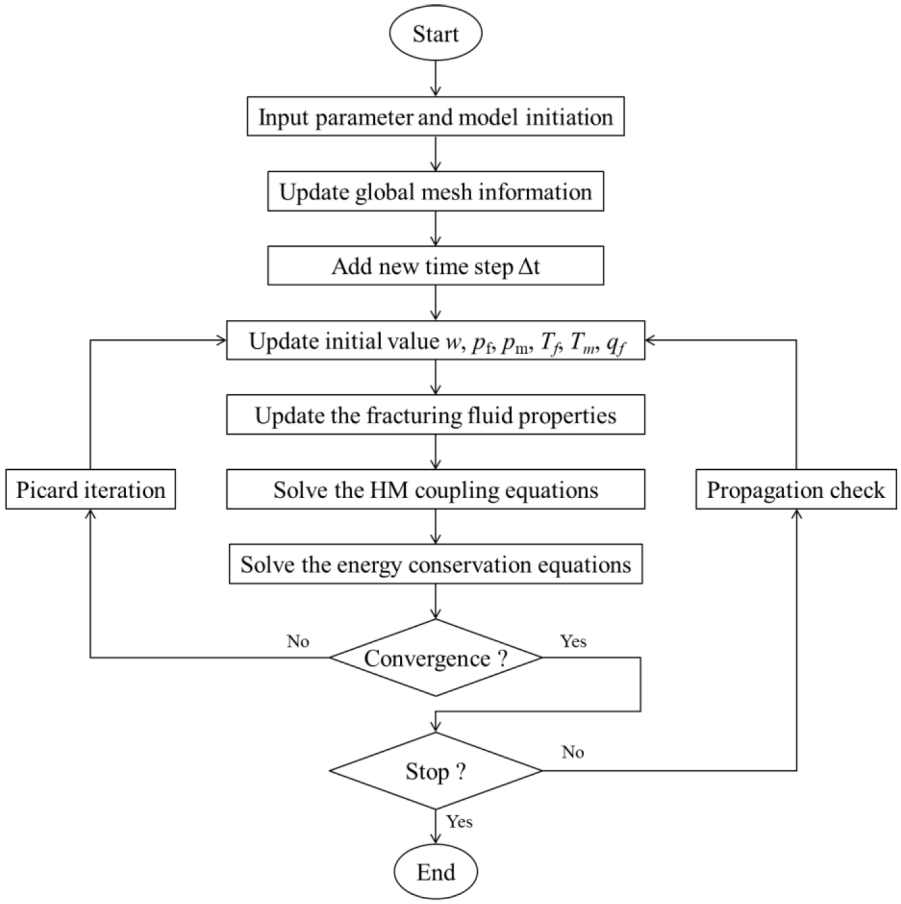


Figure 1—Computational framework of the coupled thermo-hydro-mechanical model

Results and Discussions

Model Verification

Induced stress. The induced stresses generated by the rock deformation can significantly influence the competitive propagation among multiple HFs. To validate that the fracture propagation model developed in this study, a symmetric rectangular HF oriented along the maximum horizontal principal stress (σ_H) direction is assumed to exist within the reservoir plane (Fig. 2), with uniform net pressure within the HF. The numerical solution for induced stresses can be evaluated by the introduced 3D DDM. According to linear elastic theory, the analytical solution for induced stresses generated by HF under plane strain conditions is given by (Ge and Ghassemi, 2008):

$$
\begin{aligned}
\sigma_{xx} &= p_{net}\left[\frac{r}{L}\left(\frac{L^2}{r_1 r_2}\right)^{3/2}\sin\theta\sin\frac{3}{2}\left(\theta_1+\theta_2\right)\right] + p_{net}\left[\frac{r}{\left(r_1 r_2\right)}\cos\left(\theta-\frac{1}{2}\left(\theta_1+\theta_2\right)\right)-1\right]\\
\sigma_{yy} &= -p_{net}\left[\frac{r}{L}\left(\frac{L^2}{r_1 r_2}\right)^{3/2}\sin\theta\sin\frac{3}{2}\left(\theta_1+\theta_2\right)\right] + p_{net}\left[\frac{r}{\left(r_1 r_2\right)}\cos\left(\theta-\frac{1}{2}\left(\theta_1+\theta_2\right)\right)-1\right]\\
\sigma_{zz} &= \nu\left(\sigma_{xx}+\sigma_{yy}\right)
\end{aligned} \tag{31}
$$

where p is the net pressure within the HF; v denotes the Poisson's ratio; r_1, r_2 and r are distances from a point in the plane to the upper vertex, lower vertex, and midpoint of the HF, respectively; θ_1, θ_2 and θ are angles between a point in the plane and the lines connecting it to the upper vertex, lower vertex, and midpoint of the HF, respectively; σ_{xx}, σ_{yy} and σ_{zz} are normal induced stresses along the x, y, and z directions of the reservoir.

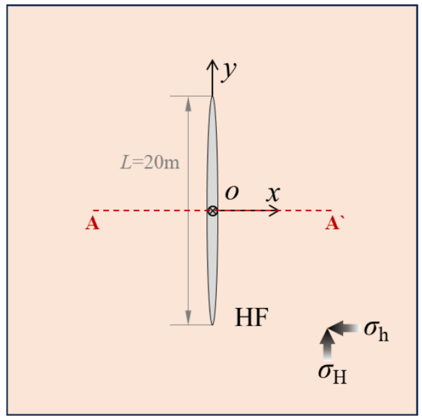
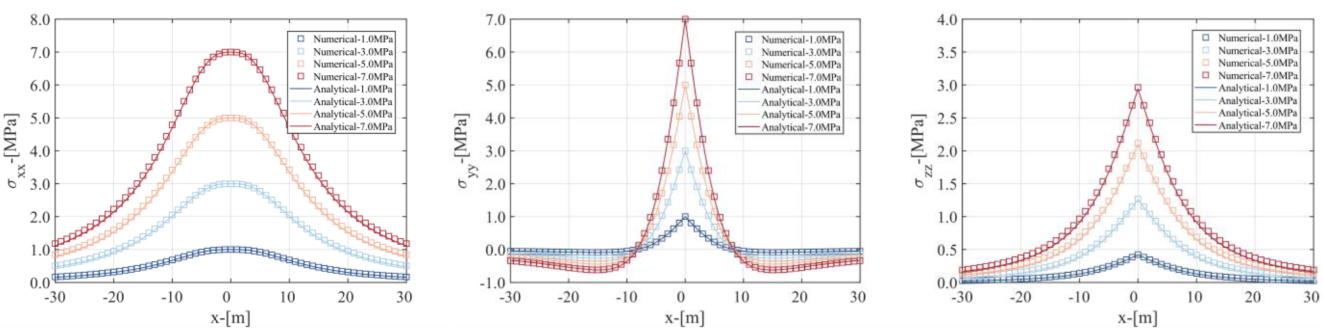


Figure 2—Schematic of hydraulic fracture

The input parameters are summarized in Table. 1. Fig. 3 illustrates the induced stresses distribution along line A-A′ (Fig. 2) under different net pressure conditions within the fractures. The numerical results obtained in this study demonstrate a good agreement with the analytical solutions, thereby validating that the developed numerical model here can effectively describe the stress interference effects among HFs.

Table 1—Input parameters of the validation case.

Parameters	Value	Unit
Youn’s Modulus	GPa	23.0
Poisson’s ratio	-	0.21
Net pressure	MPa	1.0, 3.0, 5.0, 7.0
HF length	m	20



(a) Induced stress σ_{xx} (b) Induced stress σ_{yy} (c) Induced stress σ_{zz}

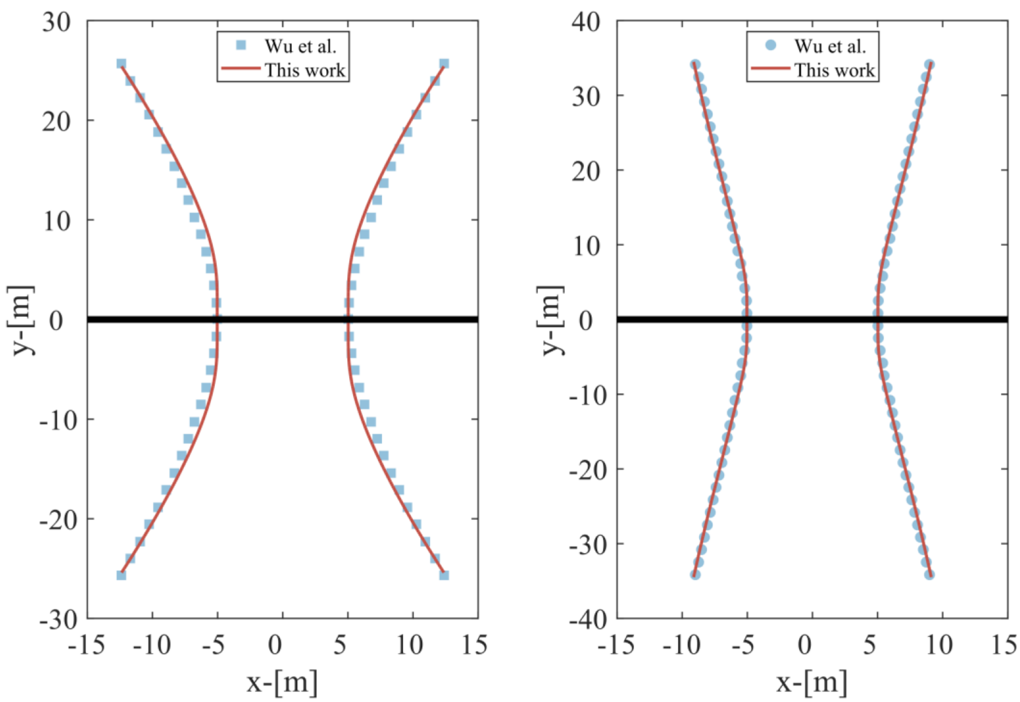
Figure 3—Results of the induced stress distribution along line A-A′ within the reservoir plane

Multi-cluster staged fracturing. Multi-cluster fracturing in horizontal wells induces stress shadow effects that alter the geo-stress around the artificial fractures, causing propagation path deviation from initial perforation direction. To validate the proposed multi-fracture propagation model, we adopted Wu et al.'s (2015) simulation scenario, comparing trajectories with and without stress differentials. Table 2 details the input parameters. To ensure that the simulation conditions are as consistent as possible, the model presented in this study limits the height of HFs to not exceed the reservoir thickness and excludes the effects of the temperature field.

Table 2—Input parameters of the validation case.

Parameters	Unit	Value
Minimum horizontal principal stress	MPa	46.70
Horizontal stress differentials	MPa	0.90
Young's modulus	GPa	29.993
Poisson's ratio	-	0.35
Injection rate	m^3/min	6.36
Fluid Viscosity	mPa·s	1.0
Fracture height	m	120.09
Cluster spacing	m	10.0

Fig. 4. shows a comparison of the simulation results from the non-planar 3D model proposed in this study with those from Wu et al. (2015). The stress interference among fractures induces progressive deviation of fracture trajectories. Moreover, the presence of horizontal stress differentials mitigates the extent of this deviation. Since our model accounts for vertical propagation of the HF, the degree of fracture deviation near the horizontal wellbore is less than that observed in the 2D model. Overall, the fracture trajectories from our model demonstrate a good correlation with those obtained by Wu and Olson.



(a) Isotropic case (b) Anisotropic case

Figure 4—Comparison of the HF propagation trajectories

Induced stresses analysis

In this subsection, the influence of induced stresses resulting from changes in reservoir pore pressure and temperature differentials on the fracture propagation during hydraulic fracturing is analyzed. Three scenarios are designed: (a) no induced stress effects; (b) only the poroelastic stress; and (c) the poroelastic stress and thermal stress. Figure 5 depicts a HF initiated along the direction of σ_H within the reservoir, with its injection point centered at (0,0,0). Corresponding input parameters are cataloged in Table 3.

Table 3—Input parameters for induced stresses analysis

Parameters	Unit	Value
Minimum horizontal principal stress	MPa	33.1
Reservoir pore pressure	MPa	24.0
Reservoir temperature	K	353.15
Reservoir thickness	m	45.0
Yang's modulus	GPa	24.5
Poison's ratio	-	0.22
Fracture toughness	$MPa \cdot m^{0.5}$	5
Stress difference in upper layer	MPa	2.0
Stress difference in lower layer	MPa	3.0
Reservoir permeability	mD	0.1
Reservoir porosity	-	0.06
Injection rate	m^3/min	3.0
Injection time	min	10
Temperature of injection fluid	K	323.15
Rock thermal expansion coefficient	K^{-1}	3.0×10^{-7}
Rock thermal conductivity	$W/(m \cdot K)$	3.29
Rock specific heat capacity	$J/(kg \cdot K)$	1000
Rock density	kg/m^3	2650

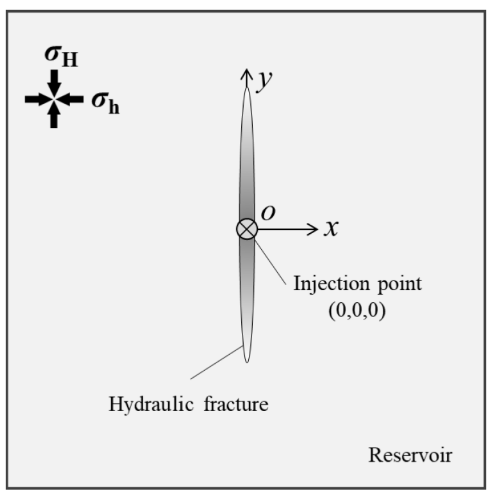
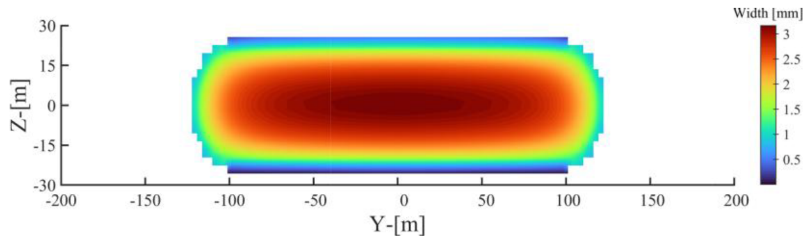
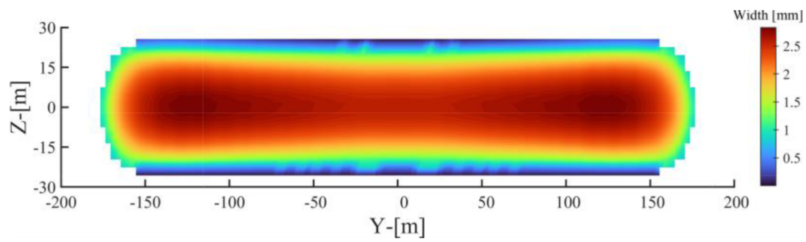


Figure 5—Schematic of the hydraulic fracturing in 2D plane

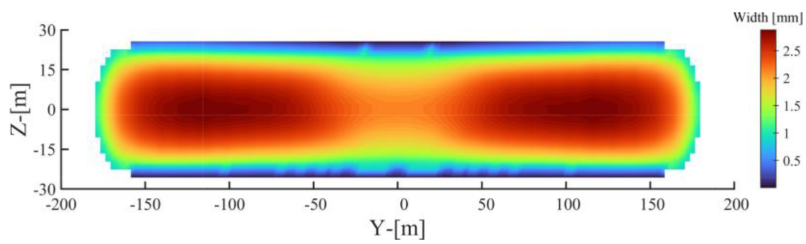
Fig. 6. shows the simulation results of 3D propagation trajectories and geometries of hydraulic fracture under different induced stresses conditions. It can be observed that when the effects of induced stresses caused by temperature differentials and fluid loss are not taken into account, the hydraulic fracture exhibits a shorter length and a larger width, with the width progressively decreasing from the injection point to the fracture front. In contrast, when additional induced stress effects are considered, the length of the hydraulic fracture increases, accompanied by a change in the distribution of fracture width.



(a) No induced stress



(b) Only poroelastic stress



(c) Poroelastic stress and thermal stress

Figure 6—The geometries of hydraulic fracture for different scenarios.

Fig. 7 displays the distribution of fracture widths along the y-axis at the end of injection. It is evident that when the effects of additional induced stresses are not taken into consideration, the fracture width achieves its maximum value at the injection point. However, when the influence of fluid loss is factored in, the substantial fluid loss occurring leads to increased additional induced stress. This results in a reduction of net pressure within the fracture near the injection point, thereby causing a decrease in the hydraulic fracture width. When the effects of both leak-off and temperature changes are considered concurrently, the injection of low-temperature fluid induces deformation of reservoir rock with higher temperature. This deformation generates additional thermal stress that influences the stress distribution acting on the hydraulic fracture surfaces, resulting in a further reduction of the fracture width near the injection point. Fig. 8. illustrates the variation curves of fluid pressure at the hydraulic fracture injection point over time. It is evident that both fluid loss of the fracturing fluid and the thermal stress induced by temperature increase the normal stresses acting on the fracture surfaces, thereby requiring a greater fluid driving force for fracture growth.

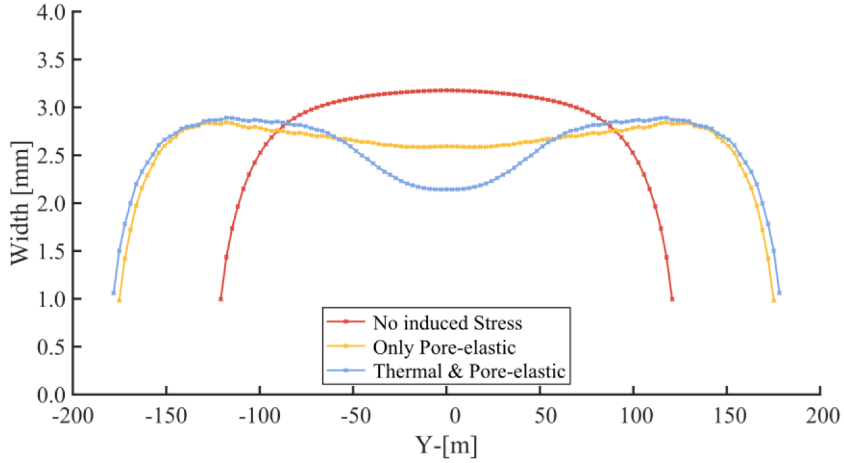


Figure 7—The distribution of fracture widths along the y-axis.

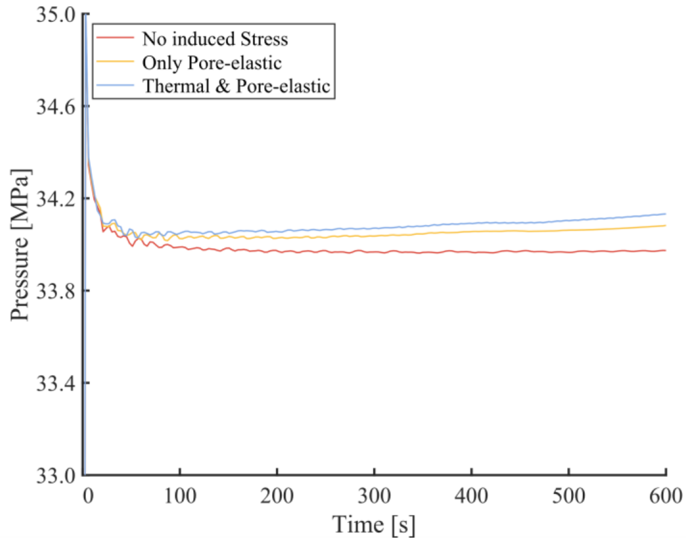


Figure 8—Fluid pressure at the injection point for different scenarios.

Comparation of different fracturing techniques

This subsection applies a validated THM-coupled propagation model to compare conventional hydraulic and SC-CO_2 fracturing, revealing the differences in artificial fracture morphologies and variations of bottom hole pressure. As depicted in Fig. 9, two parallel hydraulic fractures located in the horizontal well, with their initial propagation direction aligned with the orientation of the σ_H. The input parameters are tabulated in Table 2, and the reservoir rock parameters are aligned with those presented in Table 4.

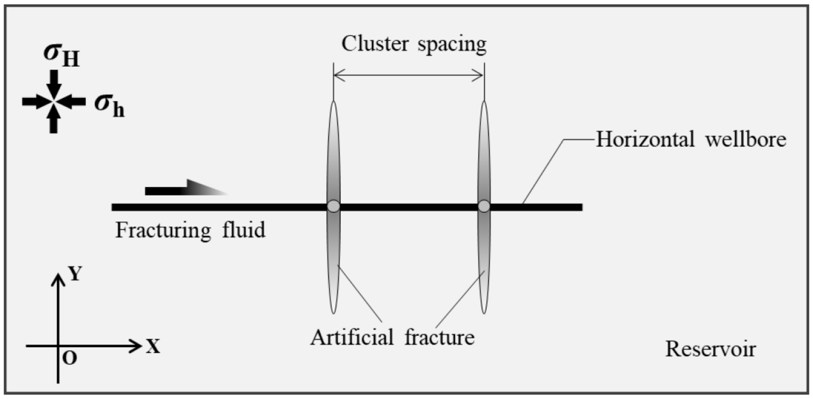


Figure 9—Schematic of two parallel artificial fractures in a horizontal wellbore

Table 4—Input parameters for comparison

Parameters	Unit	Value
Minimum horizontal principal stress	MPa	30.1
Horizontal principal difference	MPa	2.0
Reservoir pore pressure	MPa	19.0
Reservoir temperature	K	353.15
Reservoir thickness	m	40.0
Yang's modulus	GPa	24.5
Poison's ratio	-	0.22
Fracture toughness	$MPa \cdot m^{0.5}$	5
Stress difference in upper layer	MPa	2.0
Stress difference in lower layer	MPa	3.0
Reservoir permeability	mD	0.04
Reservoir porosity	-	0.06
Cluster number	Num	2
Cluster spacing	m	15
Perforation number	Num/Cluster	10
Perforation diameter	mm	12
Wellbore inner diameter	mm	104.8
Injection rate	m^3/min	3.0
Injection time	min	10
Temperature of injection fluid	K	313.15

Fig. 10. depicts the fracture geometries at the end of pumping. It is evident that, influenced by stress interference between fractures, the direction of fracture propagation exhibits a deviation, which in turn impacts the morphologies of the fractures. Given that the viscosity of water exceeds that of SC-CO_2, the volume of water lost during filtration is lower under comparable reservoir conditions. Consequently, the artificial fractures generated through hydraulic fracturing exhibit greater widths. Fig. 11. shows the comparison of fracture propagation trajectories for hydraulic fracturing and SC-CO_2 within the XOY planar. It is evident that, under identical reservoir and injection conditions, the fracture lengths resulting from SC-CO_2 fracturing are obviously shorter than those achieved through hydraulic fracturing. This difference is primarily due to the lower viscosity and higher diffusion rate associated with SC-CO_2. Additionally,

the fractures generated by SC-CO_2 fracturing demonstrate a slightly lesser degree of deflection under the influence of stress interference compared to those generated by hydraulic fracturing. This phenomenon can be attributed to the substantial loss of CO_2 into the reservoir, resulting in changes to the pore pressure that alter the stress conditions around the fractured region. Furthermore, simulation results indicate that, under equivalent conditions, the extent of stress interference during hydraulic fracturing is greater than that observed during SC-CO_2 fracturing.

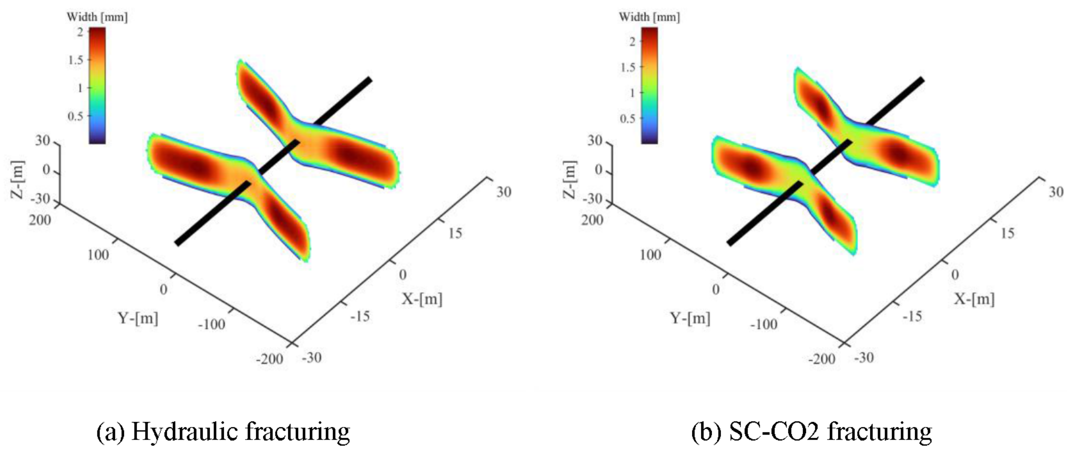


Figure 10—Comparison of artificial geometries for different fracturing techniques.

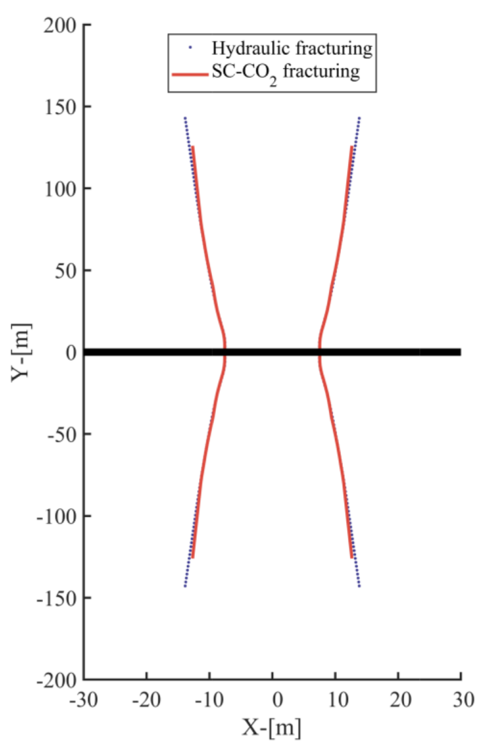


Figure 11—Comparison of facture trajectories in XOY plane.

Fig. 12. illustrates the temporal variation of bottom-hole pressure during the fracturing process. In comparison to hydraulic fracturing, SC-CO_2 fracturing is characterized by considerable fluctuations in bottom-hole pressure, which manifest as pressure increases and releases corresponding to the stop and propagation of fractures.

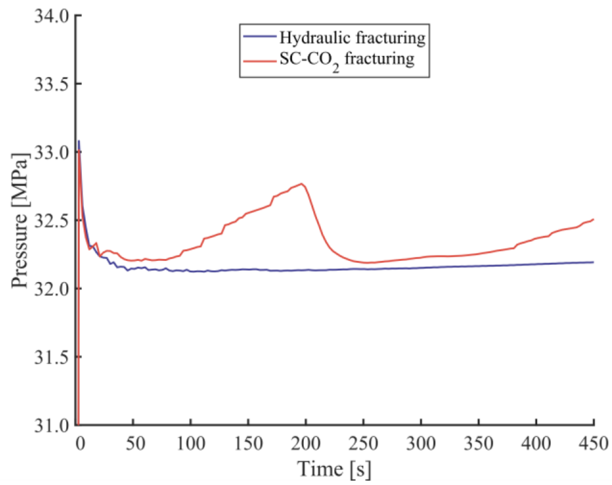


Figure 12—Variations in bottom hole pressure under different fracturing methods.

Sensitivity analysis

This section examines injection rate and cluster spacing effects on multiple fracture growth during SC-CO_2 fracturing. Fig.13 details the horizontal well multi-cluster configuration. Each artificial fracture initiates propagation parallel to the σ_H orientation. Input parameters are listed in Table 5. The reservoir rock parameters and perforation parameters are consistent with the parameters detailed Table 5.

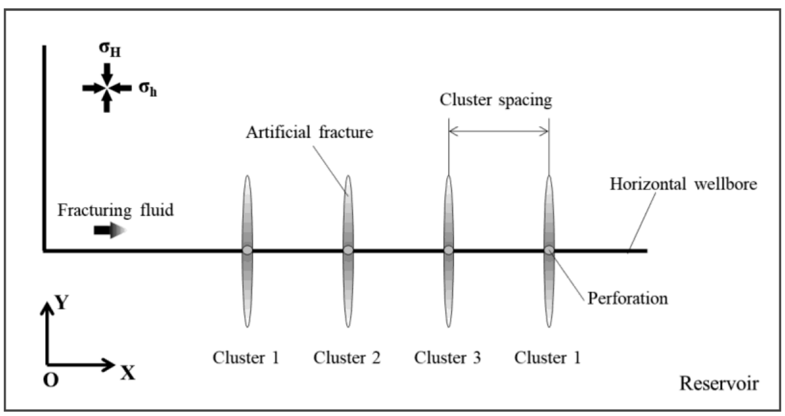


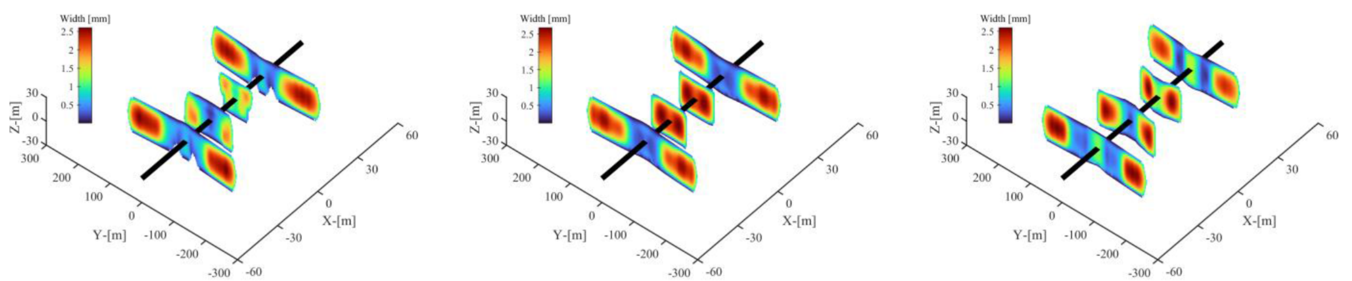
Figure 13—Multi-cluster SC-CO_2 fracturing in a horizontal well

Table 5—Input parameters for sensitivity analysis

Parameters	Unit	Value
Minimum horizontal principal stress	MPa	30.1
Horizontal principal difference	MPa	4.0
Reservoir pore pressure	MPa	19.0
Reservoir temperature	K	353.15
Reservoir thickness	m	40.0
Yang's modulus	GPa	24.5
Poison's ratio	-	0.22
Fracture toughness	$MPa \cdot m^{0.5}$	5
Stress difference in upper layer	MPa	2.0
Stress difference in lower layer	MPa	3.0
Reservoir permeability	mD	0.04

Parameters	Unit	Value
Reservoir porosity	-	0.06
Cluster number	Num	4
Injection rate	m^3/min	5.0
Injection time	min	8
Temperature of injection fluid	K	303.15

Cluster spacing. The appropriate design of cluster spacing is pivotal for improving the uniform growth of multiple fractures in horizontal well fracturing treatments. Fig. 14. depicts the fracture trajectories and width distributions for each perforation cluster at different cluster spacings. Stress interference among fractures markedly reduces the fracture length and width. Furthermore, taking into account the effects of additional induced stressed and fracture deflection, the fracture widths near the horizontal wellbore are smaller, exhibiting a trend of initial increase followed by a decrease as the distance from the horizontal wellbore increases. As cluster spacing is increased, the effects of stress interference diminish, leading to enhancements in the fracture widths and lengths of the 2nd and 3rd clusters. This finding indicates that a larger cluster spacing is more favorable for achieving uniform propagation of multiple fractures during SC-CO_2 fracturing.



(a) 16m (b) 20m (c) 24m

Figure 14—Fracture geometries of SC-CO2 fracturing under different cluster spacing

Fig. 15. shows the temporal variation of bottom hole pressure corresponding to different cluster spacings. The properties of CO_2 are highly sensitive to fluctuations in temperature and pressure, which results in significant fluctuation in the bottom hole pressure during SC-CO_2 fracturing. The combined effects of fluid loss, temperature differentials, and stress interference lead to a progressive intensification of stress interference as fracturing continues, causing the bottom hole pressure to exhibit a rising trend over time. Given that stress interference is inversely related to cluster spacing, a smaller cluster spacing results in increased resistance to fracture propagation, thereby resulting in higher bottom hole pressures.

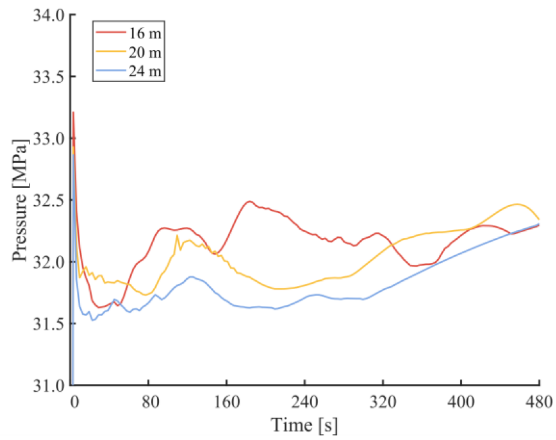
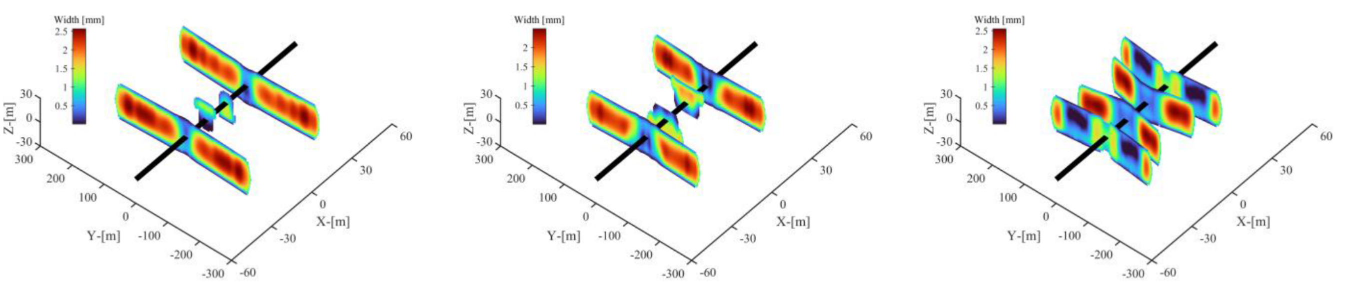


Figure 15—Variations in bottom hole pressure under distinct cluster spacings.

Injection rate. We examine how injection rate affects fracture morphology and fluid pressure during $SC\text{-}CO_2$ fracturing. Simulations employ three different injection rates—4.0, 5.0, and 6.0 m^3/min—while maintaining constant injected fluid volume. The fracture geometries (Fig. 16.) demonstrates that injection rate has a significant impact on $SC\text{-}CO_2$ fracturing. Under the conditions of identical injection volume and cluster spacing, a lower injection rate results in marked non-uniformity in the growth of multiple fractures. On the one hand, a reduced injection rate generates insufficient driven pressure, hindering effective fracture propagation at the middle perforation clusters (cluster 2 and cluster 3). Moreover, once the fractures located at the sideway clusters have sufficiently extended, the stress interference among these fractures intensifies, further aggravating the uneven propagation of multiple fractures in the horizontal well. With an increase in pumping rate, the fluid pressure within the fractures increases, thereby facilitating a more uniform growth of the multiple fractures. Notably, when the injection rate reaches 6 m^3/min, the discrepancies in fracture lengths and widths among different clusters are significantly diminished.



(a) $4m^3/min$ (b) $5m^3/min$ (c) $6m^3/min$

Figure 16—Fracture geometries of $SC\text{-}CO_2$ fracturing under different injection rate

Fig. 17. shows the variations of bottom hole pressure during $SC\text{-}CO_2$ under different injection rates. At the early stage of fracture propagation ($t < 50s$), an increase in injection rate can accelerate the velocity of fracture propagation, subsequently resulting in a rapid decline in bottom hole pressure. Under the combined influences of stress interference and additional induced stresses, the bottom hole fluid pressure may rise when fracture extension encounters resistance. Conversely, when the pressure within the artificial fractures satisfies the conditions for extension, dynamic fracture growth leads to an obvious reduction in bottom hole pressure. Notably, higher injection rate is associated with shorter durations for the bottom hole pressure to transition from an increasing to a decreasing trend.

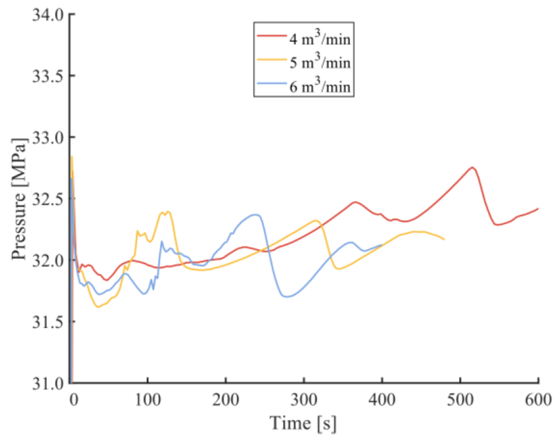


Figure 17—Variations in bottom hole pressure under distinct injection rates.

Conclusions

In this work, an integrated THM coupled numerical model is developed to investigate three-dimensional fracture propagation behaviors under SC-CO_2 fracturing, incorporating thermal and poroelastic effects. According to numerical simulation and analysis, concluding remarks are given as follows:

1. The poroelastic stress resulting from the filtration of fracturing fluid, combined with thermal stress due to the temperature differential between the fracturing fluid and the reservoir rock, alters the combined stress acting on the surfaces of hydraulic fracture. This stress change narrows fracture widths in the regions close to the wellbore while elevating the bottom hole pressures
2. Compared with the conventional hydraulic fracturing, the fractures generated by SC-CO_2 fracturing exhibit reduced fracture widths and lengths, and the bottom hole fluid pressure experiences significant fluctuations throughout the fracture propagation process due to the heightened diffusivity of SC-CO_2.
3. Fracture geometries are markedly influenced by cluster spacing and injection rate during SC-CO_2 fracturing. Increasing both cluster spacing and injection rate can mitigate stress interference and increase fluid driving pressure, thereby promoting the effective propagation of suppressed fractures and improving the overall performance of reservoir stimulation.

Reference

BennionD.B., ThomasF.B.Formation damage issues impacting the productivity of low-permeability, low initial water-saturation gas producing formations[C]. 2005: 240–247.

CollinM., RowcliffeD.Analysis and prediction of thermal shock in brittle materials[J]. *Acta Materialia*, 2000, **48**(8): 1655–1665.

DontsovE.V.An efficient computation of leak-off induced poroelastic stress for a hydraulic fracture[J]. *Journal of the Mechanics and Physics of Solids*, 2021, **147**: 104246.

ErdoganF., SihG.C.On the crack extension in plates under plane loading and transverse shear[J]. 1963: 519–525.

GeJ., GhassemiA.Analysis of failure potential around a hydraulic fracture in jointed rock[C]. ARMA US Rock Mechanics/Geomechanics Symposium, 2008: ARMA-08.

GongL., ChenS., ZuoJ., et alPhase prediction of supercritical carbon dioxide and its application in fracturing oil wellbores[J]. *Journal of Thermal Science*, 2019, **28**: 484–493.

GuptaA.P., GuptaA., LanglinaisJ.Feasibility of supercritical carbon dioxide as a drilling fluid for deep under-balanced drilling operations[C]. SPE Annual Technical Conference and Exhibition, 2005: SPE-96992.

HeY., YangZ., JiangY., et alA full three-dimensional fracture propagation model for supercritical carbon dioxide fracturing[J]. *Energy Science & Engineering*, 2020, **8**(8): 2894–2906.

HeZ., TianS., LiG., et alThe pressurization effect of jet fracturing using supercritical carbon dioxide[J]. *Journal of Natural Gas Science and Engineering*, 2015, **27**: 842–851.

Ju, Y., Zhang, G., Wu, G., et al3D fracture network morphologies and breakdown pressure in supercritical CO_2 fracturing of tight sandstones: Effects of high temperatures and in-situ stress differentials[J]. *Energy & Fuels*, 2025, **39**(8): 3820–3833.

Kim, S., Kim, Y., Oh, S.Y., et alDirect utilization of CO_2 via methanol synthesis for natural gas fields with high CO_2 concentration[J]. *Journal of Natural Gas Science and Engineering*, 2021, **96**: 104308.

Li, Q., Li, Q., Cao, H., et alThe crack propagation behaviour of CO_2 fracturing fluid in unconventional low-permeability reservoirs: Factor analysis and mechanism revelation[J]. *Processes*, 2025, **13**(1): 159.

Li, X., He, Y., Huo, M., et alModel for calculating the temperature and pressure within the fracture during supercritical carbon dioxide fracturing[J]. *International Journal of Hydrogen Energy*, 2020, **45**(15): 8369–8384.

Li, X., Li, G., Yu, W., et alThermal effects of liquid/supercritical carbon dioxide arising from fluid expansion in fracturing[J]. *SPE Journal*, 2018, **23**(6): 2026–2040.

Lin, R., Peng, S., Zhao, J., et alMultiple hydraulic fracture propagation simulation in deep shale gas reservoir considering thermal effects[J]. *Engineering Fracture Mechanics*, 2024, **303**: 110147.

Liu, S., Zhu, W., Zhang, X., et alA breakdown pressure model for supercritical CO_2 fracturing considering fluid infiltration effect using the point stress criterion[J]. *Engineering Fracture Mechanics*, 2024, **307**: 110294.

Luo, Z., Wu, L., Zhao, L., et alNumerical study on filtration law of supercritical carbon dioxide fracturing in shale gas reservoirs[J]. *Greenhouse Gases: Science and Technology*, 2021, **11**(5): 871–886.

Middleton, R., Viswanathan, H., Currier, R., et alCO_2 as a fracturing fluid: Potential for commercial-scale shale gas production and CO_2 sequestration[J]. *Energy Procedia*, 2014, **63**: 7780–7784.

Odabaee, M., Sauret, E., Hooman, K.CFD simulation of a supercritical carbon dioxide radial-inflow turbine: Comparing the results of using real-gas equation of state and real-gas property file[J]. *Applied Mechanics and Materials*, 2016, **846**: 85–90.

Sampath, K.H.S.M., Perera, M.S.A., Ranjith, P.G., et alCH_4-CO_2 gas exchange and supercritical CO_2-based hydraulic fracturing as CBM production-accelerating techniques: A review[J]. *Journal of CO_2 Utilization*, 2017, **22**: 212–230.

Shafloot, T.A., Kim, T.W., Kovscek, A.R.Investigating fracture propagation characteristic in shale using SC-CO_2 and water with the aid of X-ray computed tomography[J]. *Journal of Natural Gas Science and Engineering*, 2021, **92**: 103736.

Song, W., Ni, H., Tang, P., et alSimulation of supercritical carbon dioxide fracturing in shale gas reservoir[J]. *Journal of Thermal Science*, 2021, **30**: 1444–1451.

Span, R., Wagner, W.A new equation of state for carbon dioxide covering the fluid region from the triple-point temperature to 1100 K at pressures up to 800 MPa[J]. *Journal of Physical and Chemical Reference Data*, 1996, **25**(6): 1509–1596.

Tang, H., Liang, H., Zhang, L., et alFully 3D simulation of hydraulic fracture propagation in naturally fractured reservoirs using displacement discontinuity method[J]. *SPE Journal*, 2022, **27**(3): 1648–1670.

Vengosh, A., Warner, N., Jackson, R., et alThe effects of shale gas exploration and hydraulic fracturing on the quality of water resources in the United States[J]. *Procedia Earth and Planetary Science*, 2013, **7**: 863–866.

Wang, H., Li, G., Zhu, B., et alKey problems and solutions in supercritical CO_2 fracturing technology[J]. *Frontiers in Energy*, 2019, **13**: 667–672.

Wang, J., Zhou, F., Zhang, L., et alStudy on reason analysis and removal solution on water-locking damage in tight sandstone reservoirs[J]. *Journal of Dispersion Science and Technology*, 2020, **41**(12): 1849–1858.

Wu, K., Olson, J.E.Simultaneous multifracture treatments: Fully coupled fluid flow and fracture mechanics for horizontal wells[J]. *SPE Journal*, 2015, **20**(2): 337–346.

Yan, H., Zhang, J., Zhou, N., et alStaged numerical simulations of supercritical CO_2 fracturing of coal seams based on the extended finite element method[J]. *Journal of Natural Gas Science and Engineering*, 2019, **65**: 275–283.

Yang, C., Yang, Z., Wang, H., et alMechanisms of non-uniform propagation of hydraulic fractures: A comprehensive numerical investigation[J]. *Engineering Analysis with Boundary Elements*, 2025, **178**: 106307.

Zhang, W., Wang, C., Guo, T., et alStudy on the cracking mechanism of hydraulic and supercritical CO_2 fracturing in hot dry rock under thermal stress[J]. *Energy*, 2021, **221**: 119886.

Zhang, X., Jeffrey, R.G.Fluid-driven nucleation and propagation of splay fractures from a permeable fault[J]. *Journal of Geophysical Research: Solid Earth*, 2016, **121**(7): 5257–5277.

Zhang, X., Zhu, W., Xu, Z., et alA review of experimental apparatus for supercritical CO_2 fracturing of shale[J]. *Journal of Petroleum Science and Engineering*, 2022, **208**: 109515.

Zhao, Z., Li, A., He, J., et alA laboratory investigation of fracture propagation induced by supercritical carbon dioxide fracturing in continental shale with interbeds[J]. *Journal of Petroleum Science and Engineering*, 2018, **166**: 739–746.

Zhuang, D., Yin, T., Li, Q., et alEffect of injection flow rate on fracture toughness during hydraulic fracturing of hot dry rock (HDR)[J]. *Engineering Fracture Mechanics*, 2022, **260**: 108207.